Niko Weaver

617-852-1905 | niko.weaver@duke.edu | linkedin.com/in/niko-weaver | nmwportfolio.vercel.app

Education

Duke University — Durham, NC
B.S.E. Mechanical Engineering; Aerospace Engineering Certificate — *Expected May 2028*

Waterford School — Sandy, UT
High School Diploma, Summa Cum Laude — *Aug. 2020 – June 2024*

Experience

President — Sep. 2024 – Present
Duke Robotics Club — *Durham, NC*

- Leading a 50 member club to build autonomous underwater robots for the annual international RoboSub Competition. Coordinating mechanical, electrical, and software integration.
- Designed an AUV frame, hydrodynamic shell, and buoyancy system, reducing simulated drag by 29% in Ansys Fluent.
- Contributed to a 7th-place overall finish at RoboSub 2025 and a 3rd-place design report.

Undergraduate Researcher — Aug. 2025 – Present
Duke University General Robotics Lab — *Durham, NC*

- Lead the electromechanical design of an underwater spherical robot and the preparation of a manuscript describing the platform.
- Designed a custom PCB in KiCad that cut electronics-enclosure volume by 55% and robot mass by 0.5 kg (17%).
- Develop locomotion and underwater communication methods for coordinated multi-robot operation.

Undergraduate Research Intern — May 2025 – Aug. 2025
University of Utah FPC Lab — *Salt Lake City, UT*

- Designed a custom 4-axis robotic arm to position test models within a constrained wind-tunnel workspace.
- Built a MATLAB/Simulink model to simulate real-time motion and automate torque and drag data acquisition and analysis.
- Optimized actuator selection and link lengths against workspace and torque constraints.

Projects

Fixed-Wing UAV | *Fusion 360, Ansys Fluent, ArduPilot* — Summer 2025 – Present

- Secured $1,100 in Duke Co-Lab grant funding; designed and built three airframe revisions, increasing simulated lift by 25%.
- Analyzed an initial flight test that exposed a thrust deficit; redesigned the airframe and propulsion system, raising predicted thrust-to-weight ratio from 0.6–0.7 to 1.4 and cutting estimated mass from 2.0 kg to 1.75 kg.

Underwater Robot Locomotion | *MuJoCo, PyTorch, Python* — Summer 2026

- Built a MuJoCo digital twin for controller iteration and sim-to-real development.
- Trained PyTorch reinforcement-learning policies and benchmarked locomotion performance against a heuristic controller.

Canard-Controlled Rocket | *Fusion 360, OpenRocket, Arduino C++* — 2023

- Designed and manufactured a 1 m rocket with independently actuated canards and an Arduino-based 3-axis PID controller.
- Implemented a 1 kHz control loop with IMU-based state estimation, apogee detection, and parachute-deployment logic; traced partial engine ignition during flight testing to an ignition-wiring fault.

Technical Skills

CAD & Simulation: Fusion 360, SolidWorks, Siemens NX, Ansys Fluent (CFD), FEA, MATLAB/Simulink, MuJoCo
Manufacturing: CNC machining, manual milling & turning, FDM/SLA 3D printing, laser cutting, CAM, G-code, GD&T
Electronics & Embedded: KiCad, PCB & circuit design, STM32, ESP32, Arduino, CAN, I2C, SPI, UART
Programming & Robotics: Python, C++, PyTorch, ArduPilot, Git
Other: Systems integration, technical writing, team leadership, Spanish biliteracy